LAST NAME: _____

Practice Quiz 09A — MATH 2055 — Fall 2026

Practice for Quiz 09

The following is closed book and closed note. Don't assume the grader will know what you mean. Use full sentences and provide complete and correct solutions.

Revised using GPT-6 Astra in the Codex harness.

1. (a) Let $f : \{1, 2, 3, 4\} \to \{a, b, c\}$ be given by
$$f(1) = a, \quad f(2) = b, \quad f(3) = b, \quad f(4) = c.$$
For $X = \{1, 3\}$, compute $f(X)$ and $f^{-1}(f(X))$. State whether $f^{-1}(f(X)) = X$.

   (b) State the geometric sum formula for $1 + r + \cdots + r^n$, including its hypotheses, and prove it by induction on n.